Daniel Lofeodo

AI engineer building reliable, monitored AI assistants and automation, with testing, traceability and human oversight built in. Bilingual (English/French).

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Personal Projects

2026-05 – Present **Latent SpaceMail: Agentic AI Research Newsletter**
Live 5 months: weekly AI briefing, evaluated and monitored on GCP.

- Built and deployed **6 agentic workflows** that retrieve, filter, summarize and email content, end-to-end.
- Integrated **APIs and data sources** (Hacker News, NewsAPI, arXiv, Hugging Face, SendGrid) with **exception handling** and fallbacks.
- **Evaluated AI reliability** on hand-labeled data: review step showed no gain (60% vs. 62%); summary checker is report-only.
- **Security-tested** against prompt injection: successful attacks fell from **10/115 to 0/115**.
- **Monitoring and support:** drift, cost and health checks, weekly status email, crash watchdog.
- Documented design decisions and postmortems; sign-in and token-based **access controls**.

2026-09 – 2026-10 **RAG With Receipts: Benchmarking RAG Accuracy, Grounding, and Latency**
Hand-built RAG over 110 wiki pages, deployed on GCP Cloud Run.

- **Source traceability:** every answer cites source, automatically checked against the cited text.
- **Declines to answer** when sources don't cover it (**5 of 5** out-of-scope tests passed).
- Evaluated on 55 hand-written questions: **96%** found the right source, **84%** answered correctly.
- Deployed on Google Cloud Run; benchmarked against Google's RAG service (**build vs. buy**).

Experience

2025-04 – 2026-01 **Machine Learning Engineer** (Fixed-Term Contract), Hydro-Québec, Montreal
Regulatory dam stress-testing under extreme floods, in simulation.

- Built a **safety-bounded PPO agent**: in bounds in **9 of 9** extreme flood scenarios, **83% less spilled water** than the legacy baseline.
- Partnered **weekly with hydrology engineers** (stakeholders) and turned their feedback into agent rules.
- **Tested reliability** of standard ML models: the best broke safety limits in 7 of 9 extreme cases.
- Automated the **structured data** pipeline (sensors, weather, dams) on Databricks/Spark: runtime **down from 45 to 10 min**.

2021-06 – 2024-08 **Software Engineer**, Creaform, Montreal, Canada
3D inspection software, 30-person team, multimillion-line codebase.

- Co-built a **corrosion-detection** product earning near seven figures a year.
- Collaborated with marketing and feature teams in **Agile**; owned the flagship **UI redesign** after one year.

Education

2024-08 – 2026-05 **M.Sc. Computer Science (Machine Learning)**
Montreal Institute for Learning Algorithms, University of Montreal, Montreal, Canada

- 4.1/4.3 GPA
- Professional master's; internship report on the Hydro-Québec RL work

2017-08 – 2021-05 **B.Eng. Mechanical Engineering**
Concordia University, Montreal, Canada

- Dean's List (2021)

Technical Skills

LLM / Agentic:
AI assistants, agentic workflows, LLM APIs (Claude, OpenAI, Gemini), tool use, RAG, LangGraph, LangSmith

Reliability / Controls:
Golden datasets, LLM-as-a-judge, source traceability, prompt-injection testing, drift monitoring, access controls

Automation / Integration:
Python, SQL, REST APIs, structured and unstructured data, FastAPI, Firebase

Cloud / Data:
GCP (Cloud Run, Pub/Sub, Firestore), Azure, AWS, Databricks, Spark, Docker, GitHub Actions CI/CD

Other Languages:
C#, C++, TypeScript

Certifications

- AWS MLA (Cloud-native ML)
- Claude Code in Action

Conferences

- **Autonomous Hydroelectric Dam Operation under Extreme Flooding**, Mila Annual Conference

Languages

- English (Native)
- French (Native)
- Italian (Advanced - C1)